

RA-IZ, Russian Federation

WMO: 233310

Lat: 66.852N

Lon: 65.299E

Elev: 2920

StdP: 13.21

Time Zone: 5.00 (E05)

Period: 82-97

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-27.2	-22.8	-31.0	1.1	-25.9	-26.8	1.4	-20.9	76.6	9.6	75.6	10.2	18.4	0	1.112

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.4	66.6	57.9	62.0	54.1	58.7	52.2	59.1	64.9	56.0	60.2	53.1	57.2	8.8	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
56.4	75.4	61.3	53.8	68.7	58.9	51.0	61.7	55.5	27.1	65.2	25.0	60.1	23.1	57.2	70.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 63.3	53.8	43.9	DB	-33.3	71.5	7.5	7.4	-38.7	76.8	-43.1	81.1	-47.3	85.3	-52.8	90.7	(4)
(5)			WB	-32.0	62.5	6.3	5.0	-36.5	66.1	-40.2	69.0	-43.7	71.8	-48.2	75.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	19.3	-0.7	-0.1	9.4	12.5	22.1	38.7	48.4	42.4	32.3	18.0	6.3	0.2		
	DBStd	19.69	11.22	11.89	8.71	12.97	9.73	11.47	9.32	7.52	7.34	10.04	10.68	10.80		
	HDD50	11367	1571	1402	1258	1125	867	364	146	256	531	992	1310	1545		
	HDD65	16704	2036	1822	1723	1575	1330	789	520	702	980	1457	1760	2010		
	CDD50	144	0	0	0	0	2	25	97	19	1	0	0	0		
	CDD65	6	0	0	0	0	0	0	6	0	0	0	0	0		
	CDH74	8	0	0	0	0	0	0	8	0	0	0	0	0		
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0		
	WSAvg	16.3	18.2	16.4	17.8	17.5	15.9	14.2	13.0	13.1	15.5	18.8	16.8	18.2		
	PrecAvg	34.4	2.2	1.9	2.7	2.8	2.8	3.1	3.1	3.4	2.8	4.1	3.0	2.5		
(15) Precipitation	PrecMax	40.8	3.5	3.1	5.8	5.7	4.5	4.6	6.3	6.7	5.7	5.9	5.1	3.8		
	PrecMin	29.8	1.5	0.7	0.8	1.1	1.2	1.3	0.6	1.5	0.9	1.5	1.9	1.5		
	PrecStd	2.4	0.5	0.6	0.8	0.9	0.9	0.9	1.3	1.2	1.0	0.9	0.7	0.6		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	23.1	23.9	27.2	46.3	58.1	64.0	75.5	62.8	56.2	40.3	34.2	24.1		
		MCWB	21.0	21.3	24.6	45.4	48.5	54.7	62.3	53.1	52.6	36.6	31.9	22.6		
	2%	DB	20.3	21.4	24.7	37.0	46.4	60.8	70.9	58.5	49.0	37.2	25.3	21.4		
		MCWB	18.9	19.6	23.1	33.9	42.3	52.1	61.2	51.9	45.5	35.4	24.0	20.2		
	5%	DB	17.4	18.7	22.7	33.8	39.6	57.4	65.9	55.6	45.7	34.4	23.0	18.8		
		MCWB	16.3	17.2	21.5	30.8	36.2	50.1	57.6	49.9	43.6	33.3	21.9	17.8		
	10%	DB	14.3	15.3	20.6	30.5	35.3	54.1	62.0	52.7	42.6	31.3	21.0	15.6		
		MCWB	13.3	14.0	19.4	28.3	32.3	47.6	54.5	48.9	40.8	30.7	20.3	14.6		
	0.4%	WB	21.8	22.7	25.6	45.2	49.3	57.6	65.1	56.8	53.1	38.1	30.7	22.8		
		MCDB	22.5	23.1	26.4	46.4	55.8	61.6	72.0	59.8	56.4	39.0	34.4	23.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	19.1	19.8	23.4	34.5	42.3	53.7	61.6	53.8	46.5	35.5	24.3	20.4		
		MCDB	20.2	20.8	24.3	36.6	45.5	57.8	69.2	56.6	48.5	36.8	25.0	21.2		
	5%	WB	16.4	17.1	21.8	31.5	37.0	51.0	58.7	51.2	43.9	33.1	22.0	17.8		
		MCDB	17.6	18.4	22.8	33.4	39.6	56.5	64.5	53.7	45.2	34.3	22.7	18.8		
	10%	WB	13.4	14.1	19.7	28.1	32.6	48.6	56.1	49.2	41.2	30.7	20.2	14.7		
		MCDB	14.2	15.3	20.5	30.4	35.0	53.4	60.5	52.0	42.1	31.5	21.0	15.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.1	10.0	8.3	10.4	9.3	10.5	9.4	7.8	6.1	6.8	8.1	9.3		
		MCDBR	13.0	13.3	8.4	10.7	14.4	13.2	12.0	11.3	10.1	10.9	8.6	10.6		
		MCWBR	12.3	12.3	7.8	8.8	12.5	9.5	8.9	8.3	9.9	8.6	8.1	10.0		
		MCDBR	13.4	13.3	8.5	9.9	13.1	12.5	11.6	9.9	9.4	9.2	8.5	10.7		
	5% WB	MCWBR	12.7	12.4	8.0	9.0	12.2	10.1	9.6	8.4	9.5	8.0	8.1	10.2		
(40) Clear-Sky Solar Irradiance	taub		0.167	0.221	0.263	0.285	0.300	0.314	0.335	0.318	0.280	0.234	0.163	N/A		
	taud		1.886	2.101	2.163	2.184	2.256	2.475	2.447	2.477	2.533	2.252	1.913	N/A		
	Ebn at Noon		70	203	252	277	285	284	273	267	254	198	66	0		
	Edh at Noon		4	16	26	34	35	30	30	26	19	13	4	0		
(44) All-Sky Solar Radiation	RadAvg		18	175	657	1278	1685	1572	1604	1008	545	178	39	0		
	RadStd		2	23	74	102	144	141	207	91	51	28	4	0		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (4 neighbors)	N/A	N/A	N/A	+2.77	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page